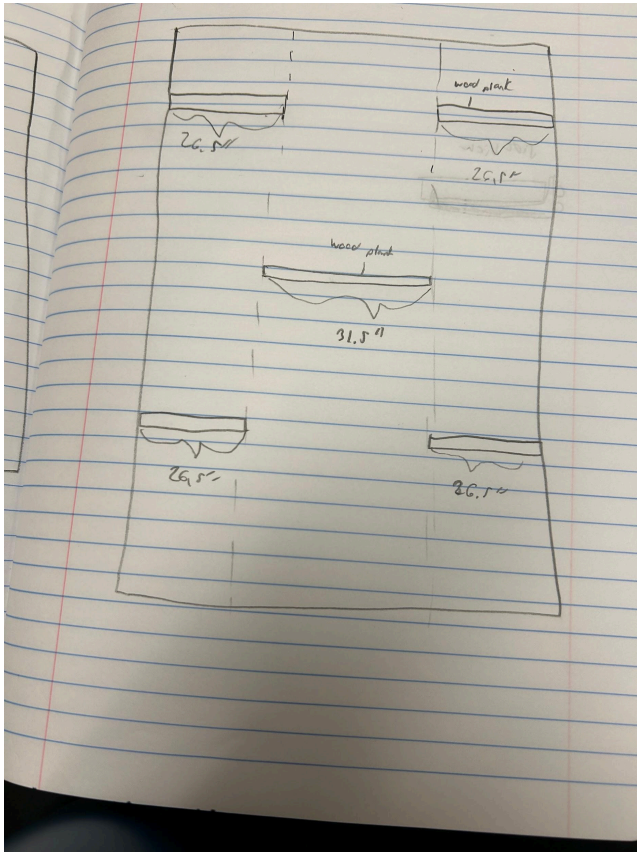


D1:

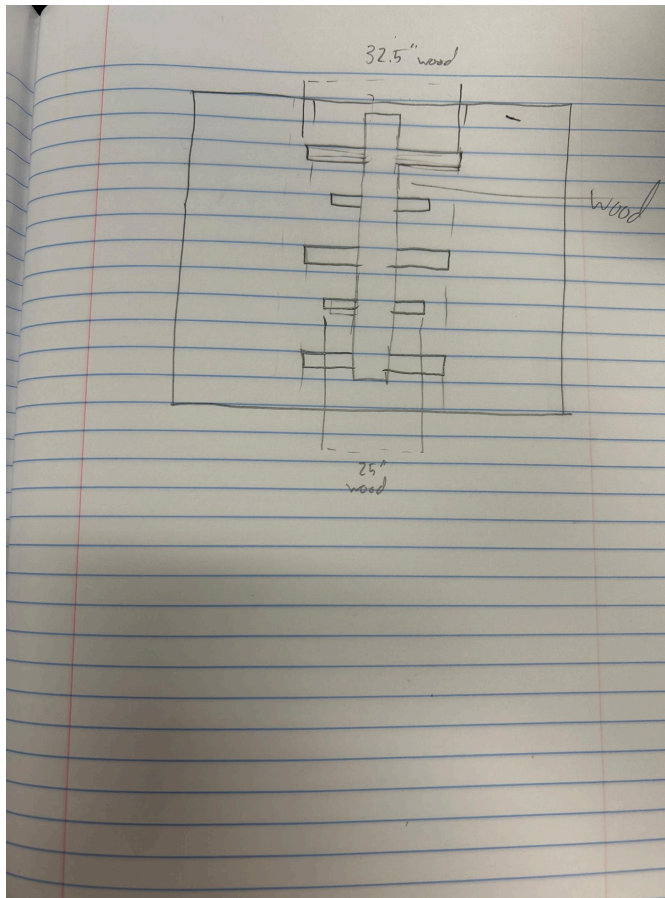
We compiled ideas that used string suspension, a staircase design, a diagonal slope design, a vertical design, and a living wood design to consider as solutions to the problem of the art showcase's shelves failing due to bending and being too heavy for the supports.

Options:

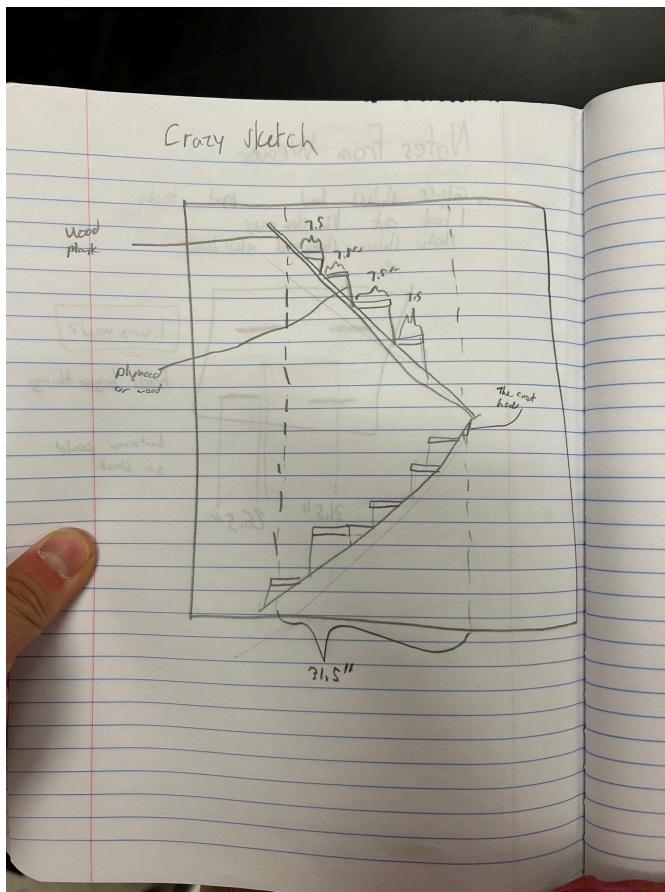
1.



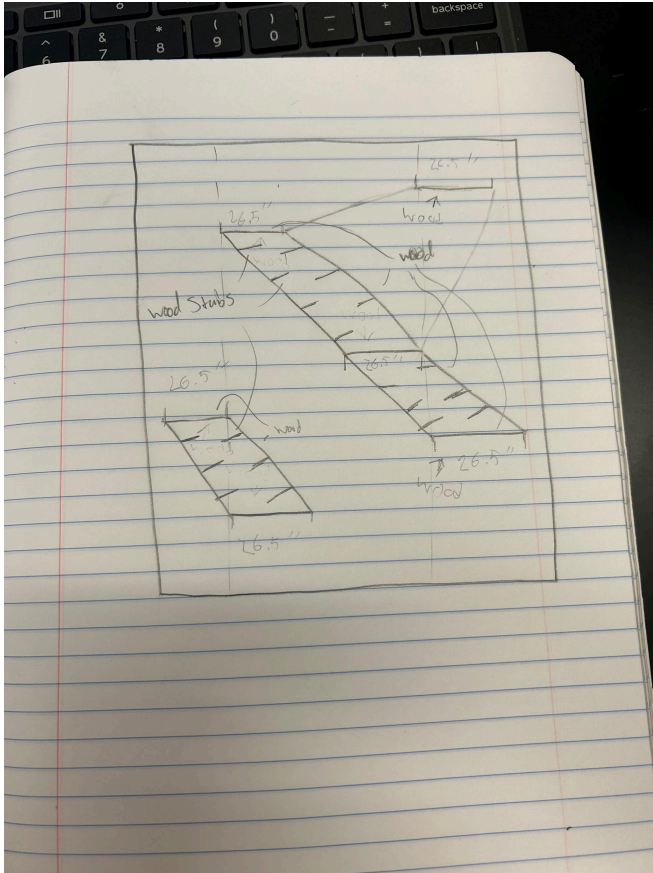
2.



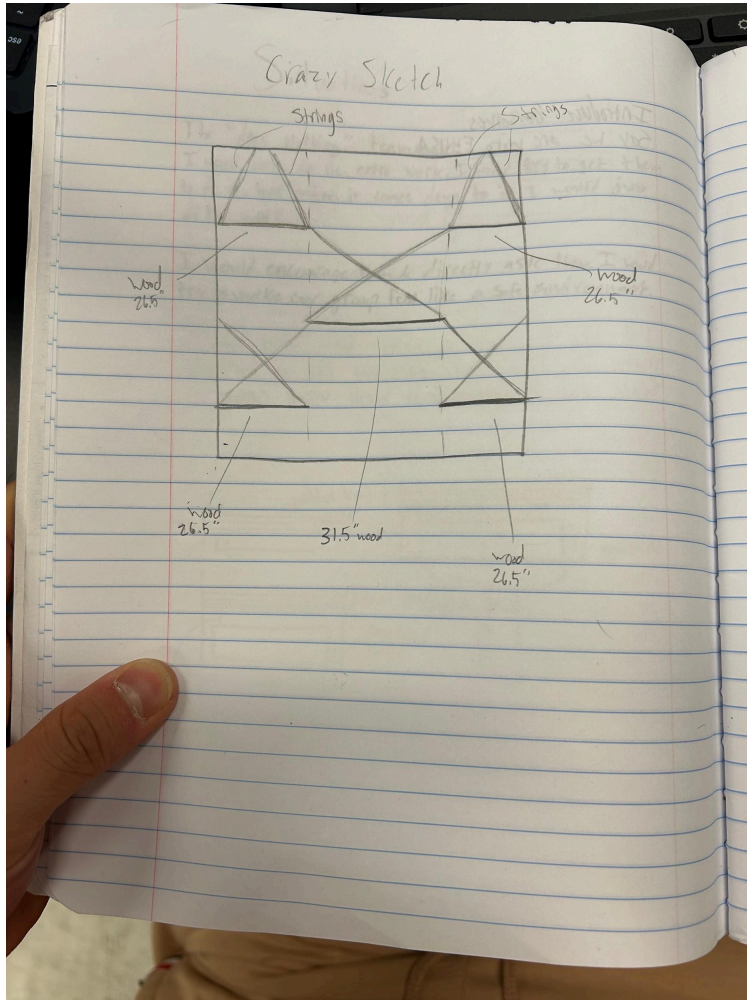
3.



4.



5.



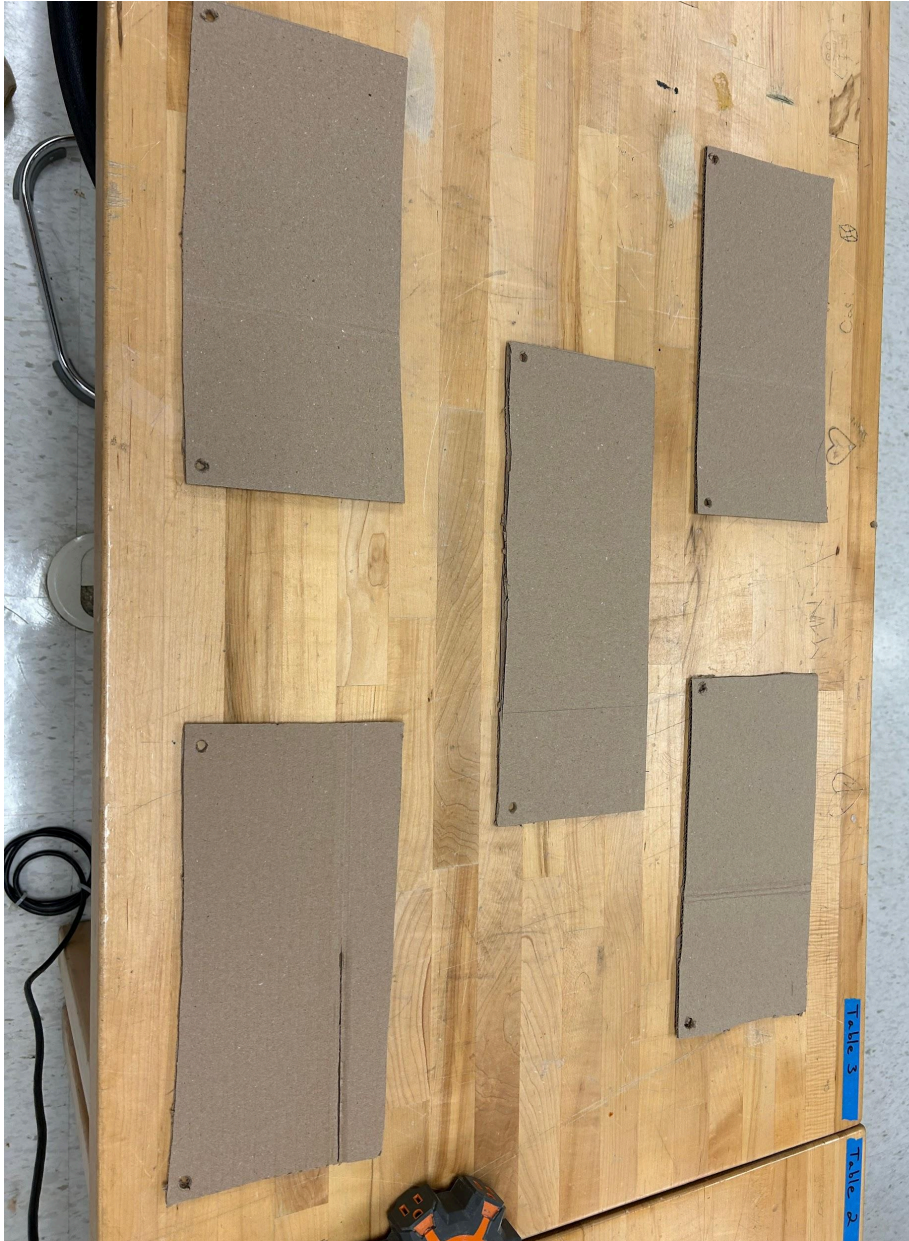
D2:

We chose a combination of option 1 and 5 because it is lightweight but sturdy and can fit in the shelf perfectly which meets the needs of Mrs. Weaver for her art showcase.

	Weight	Option 1	Option 2	Option 3	Option 4	Option 5
Hold weight of pottery	High	Yes	Yes	Maybe	Yes	Yes
Lightweight shelf	High	Yes	No	Maybe	Maybe	Yes
Row shorter than 9' 4"	High	Yes	Yes	Yes	Yes	Yes

Day 1: Materials include wood and string. We will use tape measure to find the correct measurements for where to make the cuts and holes for the strings.

Day 2-5: We will use the hand drill to make holes .5" from the side (shown in photo) for the string and the miter saw to make precise and clean cuts along the measured line.



Day 6-9: We will use the hand sander to make it smooth and nice to the touch. We will attach the string through the holes and cut it to the desired length.



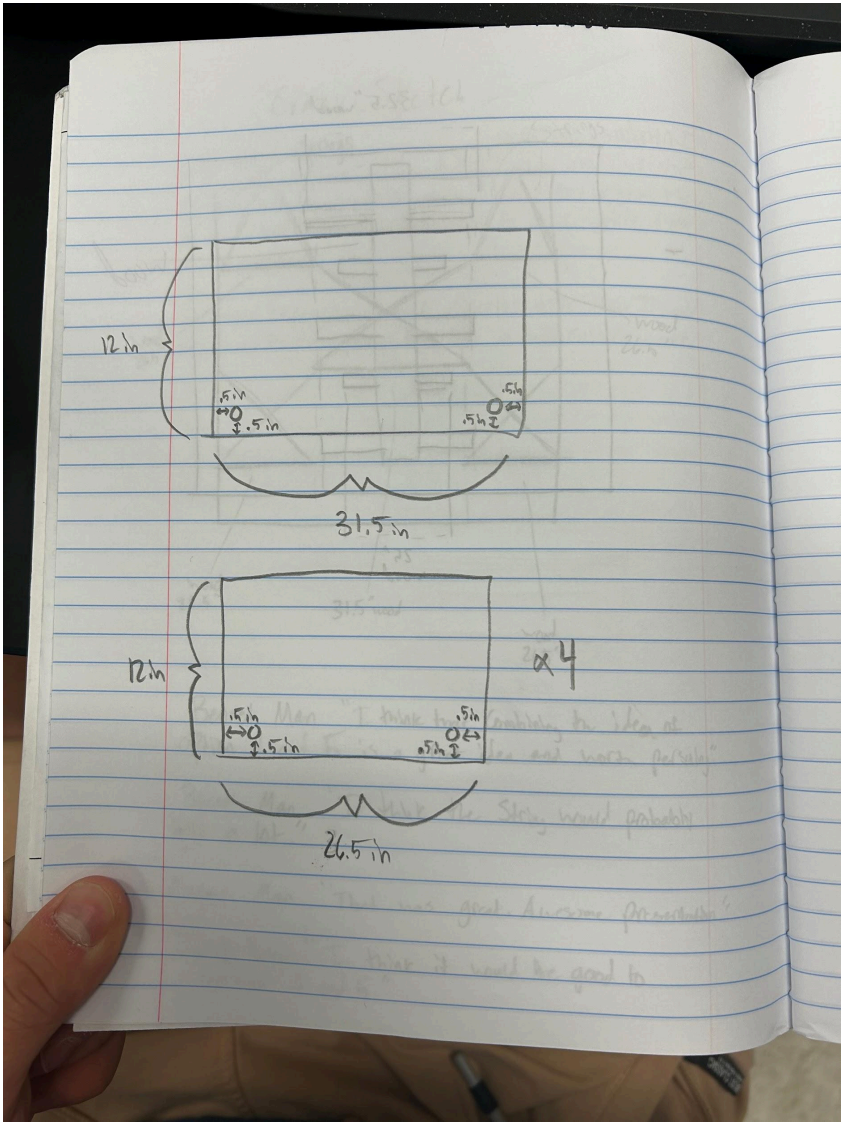
Day 10: We will find any chinks in the wood/shelf and correct them. We will make the shelf visually appealing.



D4:

Materials List:

Item Name	Quantity	Thickness	Length	Width	link
Example: 2" x 4" x 96"	1	1 ½"	14"	3 ½"	LINK
Everbilt 1/16 in x 250ft	1	1/16 in	125ft		LINK
1"x12"x3'	5	1"	12"	3'	LINK



$$7.125 + 2.208 = 4.333$$

